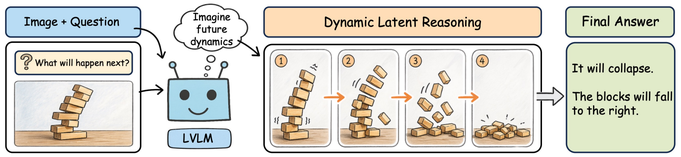
# From Seeing to Foreseeing: Unleashing LVLM Thinking in Dynamic Latent Space

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> *By imagining **future visual dynamics** in a continuous **latent space** and refining latent trajectories through **temporal supervision** and **contrastive feedback**, DLVR enables LVLMs to achieve more accurate visual reasoning.*

## Background



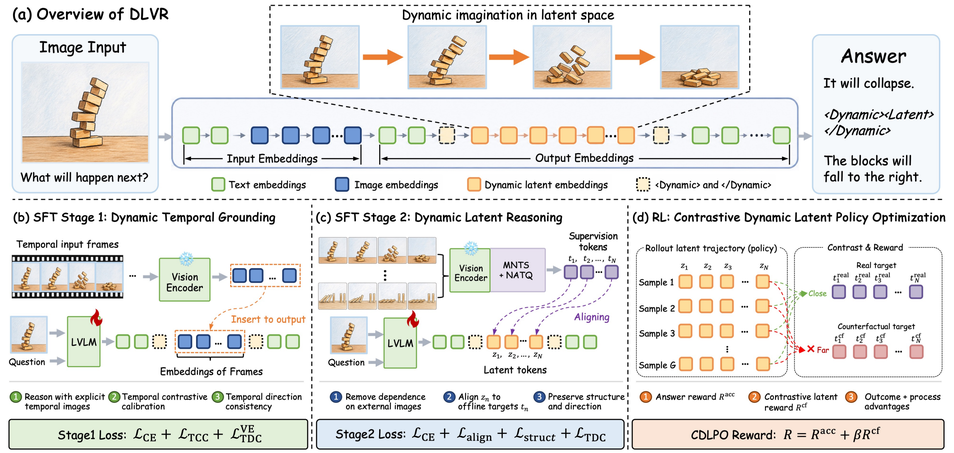

- **Think About Images** reasons over observed visual content but lacks explicit imagination of future scene evolution.
- **Think With Images** enhances reasoning through intermediate visualizations, but explicit image generation introduces additional computational costs.
- **Previous Latent Reasoning** enables implicit reasoning in continuous representations but lacks explicit grounding in temporal dynamics and future visual states.
- Our DLVR enables LVLMs to imagine future scene evolution through compact, temporally grounded latent trajectories, bridging visual understanding and dynamic reasoning.

## Contribution

- **Novel Framework**. We propose DLVR to enable LVLMs to imagine plausible future dynamics in latent space from static images.
- **Dynamic Data Construction.** We construct DLVR-SFT-100K and DLVR-RL-4K using MNTS and NATQ to provide compact supervision.
- **Progressive Training.** We introduce two-stage SFT and CDLPO to establish temporal grounding, enable latent imagination.
- **Strong Performance**. DLVR achieves improvement of dynamic benchmarks and general tasks, demonstrating effectiveness.

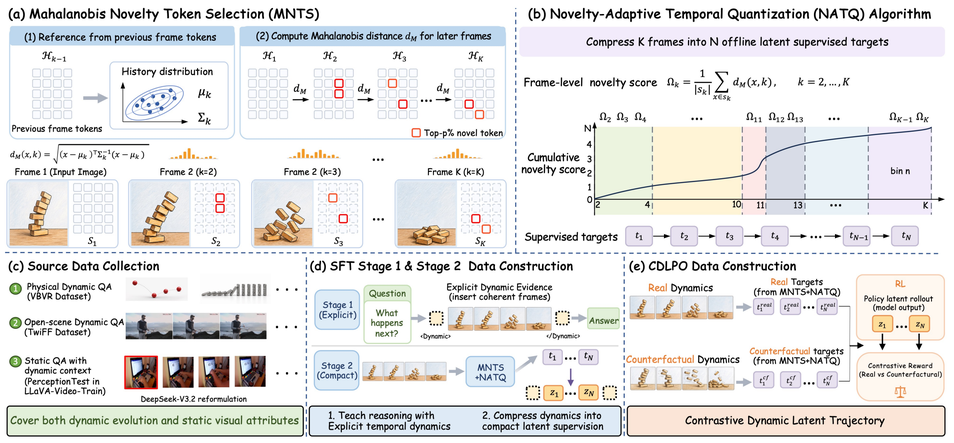
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## Method




- **Dynamic Temporal Grounding** aligns latent tokens with future visual dynamics through temporal supervision.
- **Dynamic Latent Reasoning** enables image free reasoning over latent trajectories while preserving temporal structure and direction.
- **Contrastive Dynamic Latent Policy Optimization** improves latent imagination through answer rewards and contrastive feedback from real and counterfactual futures.

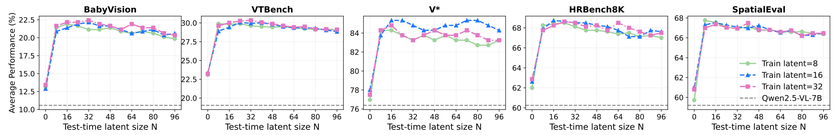
## Data Construction




- **Data Sources**: Collect diverse dynamic and static visual QA data to cover both temporal evolution and static visual understanding.
- **MNTS and NATQ**: Construct two stage SFT data with explicit temporal grounding and compact latent supervision.
- **CDLPO Data**: Build contrastive latent trajectory data from real and counterfactual futures to optimize latent imagination.

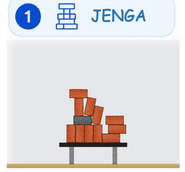
## Result

### I. Effectiveness of Latent Tokens.




- Introducing latent tokens yields substantial performance gains across benchmarks, with improvements largely saturating around 16 tokens.
- A latent size of 16 balances performance and training cost, while larger test-time budgets can further benefit dynamic reasoning tasks.

### II. Case study visualization.



**1 JENGA**

**Question:** How many of the red bricks would fall off the table if the dark grey brick wasn't there?

**Answer:** Let me imagine how the tower would evolve if that support were removed. <Dynamic><Latent></Dynamic> Without the dark grey brick, the tilted red bricks above it would lose their support and start sliding downward, which would also destabilize the upper red bricks resting on them. The right-side stack remains mostly supported by the table, while the upper leaning part collapses outward. Therefore, about three red bricks would fall off the table. The final answer is 3.



**2 RealWorld Static**

**Question:** : What's the content of the shop sign at the bottom right corner of the picture?

**Answer:** Let me imagine zooming into that region to inspect the sign more clearly. <Dynamic><Latent></Dynamic> After focusing on the bottom-right shopfront, the enlarged sign reads "International School of Languages." Therefore, the content of the shop sign is International School of Languages.

### III. Performance comparison on Dynamic-Centric and General Tasks

| Model | BabyVision | | | | | Visual Cognitive Tasks | | | VTBench | |
|---|---|---|---|---|---|---|---|---|---|---|
| | Fine-Grained | Vis. Tracking | Spatial Perc. | Vis. Pattern | Overall | Block Tower | Michotte | Astronaut | Eyeballing | Vis. Puzzles |
| Qwen2.5-VL-7B [5] | 6.13 (0.00) | 9.64 (0.00) | 15.38 (0.00) | 17.65 (0.00) | 10.57 (0.00) | 46.37 (0.00) | 16.46 (0.00) | 20.38 (0.00) | 11.42 (0.00) | 34.87 (0.00) |
| *Think With Image* | | | | | | | | | | |
| +VISER [33](NeurIPS'25) | 7.36 (1.23) | 6.02 (3.62) | 16.48 (1.10) | 23.53 (5.88) | 11.34 (0.77) | 47.32 (0.95) | 16.33 (0.13) | 19.87 (0.51) | 11.90 (0.48) | 36.08 (1.21) |
| +ICoT [26](CVPR'25) | 10.43 (4.30) | 8.43 (1.21) | 10.99 (4.39) | 21.56 (3.91) | 11.60 (1.03) | 48.91 (2.54) | 17.04 (0.58) | 20.11 (0.27) | 12.38 (0.96) | 34.27 (0.60) |
| *Latent Reasoning Method* | | | | | | | | | | |
| +CoVT [53](arXiv'25) | 11.04 (4.91) | 10.84 (1.20) | 14.29 (1.09) | 15.69 (1.96) | 12.37 (1.80) | 49.11 (2.74) | 15.80 (0.66) | 22.89 (2.51) | 13.33 (1.91) | 36.69 (1.82) |
| +LVR [36](ICLR'26) | 9.82 (3.69) | 12.05 (2.41) | 18.68 (3.30) | 17.66 (0.01) | 13.40 (2.83) | 51.07 (4.70) | 18.59 (2.13) | 22.31 (1.93) | 13.80 (2.38) | 36.89 (2.02) |
| +Monet [70](CVPR'26) | 10.42 (4.29) | 13.25 (3.61) | 15.38 (0.00) | 21.57 (3.92) | 13.91 (3.34) | 50.94 (4.57) | 17.73 (1.27) | 23.60 (3.22) | 13.91 (2.49) | 37.50 (2.63) |
| **+DLVR (Ours)** | **16.56** (10.43) | **18.07** (8.43) | **26.37** (10.99) | **33.33** (15.68) | **21.39** (10.82) | **57.79** (11.42) | **22.15** (5.69) | **30.03** (9.65) | **23.34** (11.92) | **42.39** (7.52) |

| Model | Comprehensive | | | | RealWorld | | | Spatial |
|---|---|---|---|---|---|---|---|---|
| | MMBench | MMStar | MMVet | V* | HRBench4K | HRBench8K | MME-RealWorld | SpatialEval |
| Qwen2.5-VL-7B [5] | 81.16 (0.00) | 61.18 (0.00) | 68.95 (0.00) | 76.38 (0.00) | 64.13 (0.00) | 60.25 (0.00) | 42.97 (0.00) | 59.20 (0.00) |
| *Think With Image* | | | | | | | | |
| +VISER [33](NeurIPS'25) | 82.06 (0.90) | 64.29 (3.11) | 70.28 (1.33) | 78.15 (1.77) | 65.13 (1.00) | 60.75 (0.50) | 42.40 (0.57) | 61.23 (2.03) |
| +ICoT [26](CVPR'25) | 82.18 (1.02) | 63.87 (2.69) | 70.65 (1.70) | 79.79 (3.41) | 66.62 (2.49) | 61.00 (0.75) | 43.36 (0.39) | 59.54 (0.34) |
| *Latent Reasoning Method* | | | | | | | | |
| +CoVT [53](arXiv'25) | 81.82 (0.66) | 63.75 (2.57) | 72.49 (3.54) | 78.60 (2.22) | 67.88 (3.75) | 65.13 (4.88) | 46.24 (3.27) | 62.42 (3.22) |
| +LVR [36](ICLR'26) | 81.53 (0.37) | 64.21 (3.03) | 72.85 (3.90) | 80.94 (4.56) | 68.00 (3.87) | 63.75 (3.50) | 49.66 (6.69) | 62.50 (3.30) |
| +Monet [70](CVPR'26) | 82.90 (1.74) | 62.56 (1.38) | 73.67 (4.72) | 83.11 (6.73) | 67.38 (3.25) | 64.38 (4.13) | 51.70 (8.73) | 61.94 (2.74) |
| **+DLVR (Ours)** | **86.01** (4.85) | **65.53** (4.35) | **75.83** (6.88) | **85.34** (8.96) | **74.63** (10.50) | **68.75** (8.50) | **52.53** (9.56) | **67.53** (8.33) |

- DLVR achieves SOTA performance on both dynamic centric and general benchmarks.